

- $M = p \times q$ matrix $\left\{ \begin{array}{l} M^t \text{ (Transpose of } M) \text{ is the } q \times p \\ \text{matrix with entries } (M^t)_{ij} = M_{ji} \end{array} \right.$

- Easy properties: $(M^t)^t = M$, $(M+N)^t = M^t + N^t$, $(cM)^t = cM^t$
 M, N of same size $c \in \mathbb{R}$

Also observe: $\left. \begin{array}{l} A \text{ is } p \times q \\ B \text{ is } q \times r \end{array} \right\} \Rightarrow \left. \begin{array}{l} A^t = q \times p \\ B^t = r \times q \end{array} \right\} (AB)^t = B^t A^t$

- Equation $Ax=b$ is equivalent to $x^t A^t = b^t$ (x^t, b^t row vectors)

Can solve such equations by doing column operations

- Doing a column operation is same as multiplying on the right by an elementary matrix: $A \xrightarrow[\text{cp}]{\text{column}} AE$
- Can get (reduced) column echelon form of any matrix

Take $r \times c$ matrix A , giving map $f_A: \mathbb{R}^c \rightarrow \mathbb{R}^r$
 $x \mapsto Ax$

How to find bases for these subspaces?

- (i) $\ker f_A = \text{nul } A = \{x \mid Ax=0\} \subset \mathbb{R}^c$
- (ii) $\text{im } f_A = \{Ax \mid x \in \mathbb{R}^c\}$
 $= \text{span}(\text{columns of } A) =: \text{Col}(A)$
- (iii) Similarly $\text{span}(\text{rows of } A) =: \text{Row}(A)$

(i) $\text{Nul}(A) = \{x \mid Ax=0\} = \{x \mid Rx=0\}$ where $R = \text{RREF}(A) = EA$

Example [E = product of elementary matrices, so invertible]

$A \xrightarrow[\text{red.}]{\text{row}} R = \begin{bmatrix} 0 & 1 & p & 0 & q & 0 \\ 0 & 0 & 0 & 1 & r & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

pivot variables x_2, x_4, x_6
 free variables x_1, x_3, x_5

special solutions obtained
 by setting one free variable = 1
 and the others = 0

Solutions of $Ax=0$ are

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} x_1 \\ -px_3 - qx_5 \\ x_3 \\ -rx_5 \\ x_5 \\ 0 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ -p \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 0 \\ -q \\ 0 \\ -r \\ 1 \\ 0 \end{bmatrix}$$

A basis of $\text{Nul}(A) = \text{special solutions}$ { Clearly span $\text{Nul}(A)$.
 Independent by looking
 at free variable slots }

(ii) Basis of $\text{Col}(A)$?

Natural to use the cut down strategy

Case 1 If A is in reduced echelon form, it is easy to see that the pivot columns are linearly indep & also that all other cols of A are in their span. Thus in this case a basis of A in RREF = pivot cols of A

General case Row operations change the col space of a matrix (e.g. $\begin{bmatrix} \sqrt{2} \\ \pi \end{bmatrix} \xrightarrow{\text{row}} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ have very diff col spaces)

It is good to find out linear relations among column vectors of A (so we can drop "unnecessary" columns). But that's the same as finding nontrivial solutions to $AX=0$
i.e. $\text{---} // \text{---}$ to $\text{RREF}(A)x=0$

Thus any set of columns in $\text{RREF}(A)$ is independent/dependent $\Rightarrow$ corresponding columns of A are $\text{---} // \text{---}$.

A basis for $\text{col}(A) =$ cols of A corresponding to pivot cols of $\text{RREF}(A)$

Imp remark

Nullity of $A = \text{Dim of Nul space} = \# \text{ free variables}$
(Col) Rank of A Dim of Col space = $\#$ pivot variables

Theorem $\begin{cases} \text{rank}(A) + \text{nullity}(A) = \# \text{ of columns of } A \\ \dim(\text{im } f_A) + \dim(\text{ker } f_A) = \dim(\mathbb{R}^c) \rightarrow \text{domain of } f_A \end{cases}$

(iii) Basis for $\text{Row}(A) := \text{span of row vectors of } A$

Case 1 If A is in RREF then it is easy to see that the nonzero rows form a basis, (b/c they are lin-indep)

General case arbitrary matrix A . Q. What effect do row op^n have on Row space of a matrix

$A \xrightarrow{\text{row op}^n} A'$ rows of A' are lin comb of rows of A
AND vice versa (b/c row op^n are reversible)

A basis for $\text{row}(A) = \text{nonzero rows of RREF}(A)$

Thus $\dim \text{Row}(A) = \dim \text{Col}(A) = \# \text{ pivots in RREF}(A)$

i.e. **Theorem:** $\text{row rank of } A = \text{column rank of } A \stackrel{\text{def}}{=} \text{rank of } A$

Exercises (i) How to get bases for $\text{row}(A)$ and $\text{col}(A)$ by using column operations?

(ii) $\text{Col}(A) \perp \text{Nul}(A)$ are "connected"

$\text{row}(A) = \text{Col}(A^t)$ should be similarly connected to $\text{Nul}(A^t)$

How will you find a basis for $\text{Nul}(A^t)$ using Row opⁿ on A ?

Next Back to abstract vector spaces

More examples : • Set of all $p \times q$ matrices is a v-space under usual matrix addition and scaling

(It is isomorphic to $\mathbb{R}^{pq}$)

- If X is any set then the set of functions from X to $\mathbb{R}$ is a vector space under
pointwise operations.
$$\begin{cases} (f+g)(x) := f(x) + g(x) \\ (cf)(x) := c f(x) \end{cases}$$

Added after class

I stated this observation very quickly at the end of class. I will repeat it next time, but do think about it on your own till it becomes transparent.

Q. What is the dimension of this vector space?